



ISE5001: Technology Transfer and Commercialization
1st Semester 2026/2027

Amicon Corporation (A)

(“Amicon”)

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This case (“Amicon Corporation”) will be the basis for the classroom licensing game on Day 4 of the course (Sunday 11th October) which forms a core part of Assessment Item 2 (the Negotiation Exercise). During the period between Day Two of the course and Day Four of the course, all students should carefully study the case and meet together in their assigned teams to discuss the case and to plan a negotiation strategy for their team.

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Amicon Corporation (A)

"It's five-and-a-half years since we were awarded a patent on our blood filtration technology, and so far we've not a dollar to show for it. We must decide on the best way to generate revenue from our invention or the patent will expire and our research and development effort will have been wasted," said Mr. Norman Jacobs, president of Amicon Corporation to a meeting of the senior management committee in January 1978.

In December 1972, the United States Patent Office had granted Amicon a patent on a new and superior method of separating blood plasma from whole blood. The patent claims had recently been confirmed in a two-year laboratory study conducted by the American National Red Cross, a federally chartered nonprofit organization, and the largest single blood collection organization in the United States. However, Jacobs was unsure how to turn ownership of the patent into profit for Amicon over the remaining 11¹/₂-year life of the patent. He thought perhaps Amicon should produce and market products itself, possibly find a joint-venture partner, or maybe even grant licenses to other manufacturers.

The Company

Amicon was founded in 1962 by Dr. Alan S. Michaels, a professor of chemical engineering at Massachusetts Institute of Technology and some of his former students. In 1971, Norman Jacobs, a Harvard Business School graduate who was administrative vice president and treasurer, and one of the founders, was appointed president when Dr. Michaels moved from the Northeast, where Amicon was located, to the West Coast.

Originally Amicon concentrated on performing contract research for industrial clients in the areas of chemical separation, surface chemistry, and polymer chemistry. The arrangements with clients were typically on a joint-venture basis, rather than cost-plus-fee, so that Amicon retained an ownership interest in the technology which it developed. Resulting products (or processes) outside the client's field of interest could then be manufactured (or employed) by Amicon or licenses to other companies.

¹The American National Red Cross was a 97-year-old federally chartered nonprofit organization.

This case was prepared by Dr. Noel Capon, Visiting Lecturer, with the assistance of Professor E. Raymond Corey through the cooperation of Amicon Corporation. It is intended as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

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In 1977 Amicon's profits exceeded \$500,000, on sales of over \$11 million. Over the past two years sales growth had averaged 222 per annum and Jacobs confidently expected revenues to reach \$20 million by 1980. Income statements and balance sheets for the years 1974–1977 are shown in **Exhibits 1 and 2**.

Amicon's sales and profits were derived from three basic businesses—polymer products, scientific systems, and research services and licensing. The Polymer Products Division produces and sold a range of epoxy adhesives and electrical insulating compounds to the automotive and electronics markets. By 1977 its sales of \$4.2 million were 38% of corporate sales and accounted for 39% of corporate profits. The Scientific Systems Division (SDS) produced and sold a range of membrane filters and associated equipment to medical research laboratories; by 1977 its sales of \$6.7 million were 59% of corporate sales and accounted for 64% of corporate profits. Revenues from research services and licensing were only 32 of corporate sales and represented a 3% operating loss in 1977. Sales and profit history of the business segments for 1973–1977 is shown in **Exhibit 3**

Jacobs said that whereas in Amicon's early years there had been a corporate emphasis on developing and licensing technology, growth in recent years had come almost entirely from products which Amicon manufactured and sold. While much of the growth had resulted from Amicon's own research, in the past three years Amicon had acquired a small line of related epoxy products, technology for additional membrane filters, and distribution rights in the United States and Japan for a line of British laboratory products. The majority of Amicon's sales was made in the United States, but the company was becoming increasingly active in foreign markets. By the close of 1977, over one-third of Amicon's sales were to customers located outside of the continental United States.

Background To Blood Filtration

Amicon was involved in two quite different types of blood filtration applications, the separation of blood plasma from whole blood and the removal of toxic molecules from blood. Both of these applications had been developed as an outgrowth of research on ultrafiltration membranes, the Scientific Systems Division primary product line.

SDS products were of two basic types: disposable membrane filters which were essentially thin plastic films with pores and the laboratory equipment in which the membranes were placed. The membrane filters had been developed by Amicon in the mid-1960s as a result of contract research financed in part by Dorr-Oliver Corporation. Called ultrafiltration membranes, for the first time they enabled molecules of size 10–150 Angstroms² to be filtered successfully. Thus, for instance, water and salt molecules, less than 10 Angstroms in size, could pass through an ultrafiltration membrane, but viruses, enzymes, and protein molecules which were over 10 Angstroms in size could not.

Use of ultrafiltration membranes fulfilled two functions; first, a solution containing extraneous large molecules could be cleansed of these impurities and second, protein and other biological molecules could be obtained in a more concentrated form.

Amicon's agreement with Dorr-Oliver specified that Dorr-Oliver could use the new technology to manufacture products for industrial applications, whereas Amicon was free to manufacture and sell products to the research laboratory market. Each firm paid the other royalties on the basis of 51 of membrane sales, but not on sales of the associated equipment which each of them made.

²Angstrom is a unit of length measure. One Angstrom is equivalent to 0.00000001 or 10⁻⁸ cms.

Amicon sold its ultrafiltration products to the life science and medical research laboratories of universities, hospitals, and pharmaceutical companies for a variety of research applications. Sales were made to approximately 3,000 accounts in the United States by a 10-person sales force, backed by a small internal marketing and technical service staff. In addition, Amicon employed resident salespeople in Canada, Great Britain, France, Germany, Holland, and Japan; and sold through distributors in other countries. Jacobs estimated that Amicon's sales accounted for over 75% of the world market for ultrafiltration membranes and associated laboratory equipment in research laboratories. Amicon manufactured all of its ultrafiltration membranes.

In the late 1960s, Amicon's management became dissatisfied with Dorr-Oliver's progress in developing industrial applications for ultrafiltration membranes and initiated a renegotiation of the license agreement. Under the new agreement, Amicon and Dorr-Oliver were each free to address any market it chose, each agreeing to pay the other a 5% royalty on sales.

In 1969 Amicon established a new group to design, develop, manufacture, and sell industrial ultrafiltration equipment. However, after three years of modest sales and heavy start-up losses, management decided that the company could not support this program on its own. Amicon then formed a joint venture in 1972 with Rohm & Haas, Romicon, in which Amicon held an equity interest of slightly under 20%. In forming Romicon, Amicon provided its patents, production know-how, plant, inventory and key personnel, while Rohm & Haas supplied the start-up capital. Romicon, which had its own plant and administrative offices, sold ultrafiltration membranes and associated equipment for industrial applications through a small sales force to customers in the United States and Europe. Mr. Jacobs said that industrial applications had been more difficult to develop than had been anticipated but that good progress had been made in three specific applications: paint recovery in the electrocoat paint process widely used in the manufacture of such products as automobiles and appliances, protein extraction for cheese waste, and water purification. Sales had just reached the \$2 million level, and although Romicon had yet to turn a profit, Mr. Jacobs said it had approached a breakeven level.

Blood Filtration For Toxin Removal

A notable recent technical breakthrough by Amicon was in the removal of toxins from blood, a function performed in the human body by the kidneys. The artificial kidney field was one of the fastest growing areas of medicine in the United States, Western Europe and Japan. Mr. Jacobs believed that for certain patients Amicon's system was proving superior to existing artificial kidney systems.

All artificial kidneys currently in use worked by means of a process called dialysis. In dialysis the patient's blood flowed in a continuous manner on one side of the membrane and a wash fluid flowed on the other side. During the process, toxins in the blood traveled through the membrane and were carried away in the wash fluid, while the cleansed blood was returned to the patient. Although thrice-weekly artificial kidney treatments were sustaining 100,000 patients worldwide, the treatment did not fully duplicate the natural kidney function, and had numerous undesirable side effects for most patients.

Amicon's breakthrough was the development of the *ultrafiltration* membrane which allowed blood to be filtered in a manner similar to the operation of the human kidney. When blood flowed over an ultrafiltration membrane surface, toxins in the blood passed through the membrane, while the blood, cleansed of the toxins, was retained, and could be returned to the patient. Use of the Amicon system provided a simple method of removing excess fluid from the patient as well as reducing the incidence of the undesirable side effects of hypertension and build-up of fat molecules for certain patients. Although the exact reason for these patient benefits was not known, Jacobs said a

possible explanation was that when the Amicon system was used, certain larger toxic molecules were removed to a greater extent than was possible with dialysis.

As of early 1978, the use of Amicon's ultrafiltration membranes for blood cleansing was being tested clinically to generate data on the safety and efficacy of the process for submission to the Food and Drug Administration (FDA). FDA approval was required before the Amicon device could be manufactured in the United States and sold commercially. However, interest from Western Europe, particularly Germany, had been so enthusiastic that Amicon was building a plant in Ireland to serve the European market³. Sales would be made direct to hospitals and kidney treatment centers by newly trained salespersons.

Separation Of Blood Plasma From Whole Blood

Amicon's development of filtration technology for separating blood plasma from whole blood was a direct result of solving an existing problem in ultrafiltration. The problem in ultrafiltration had been the build-up of a filter cake on the membrane surface, which reduced membrane efficiency by reducing the flow rate through the membrane. Amicon's scientists developed a method for flowing the solution to be filtered through thin channels in a direction parallel to the membrane to ensure continuous circulation on the membrane surface. This technique proved to be an effective way of solving the problem.

Having developed the parallel-flow technique in ultrafiltration, Amicon scientists investigated the possibility of employing the same technique in other types of filtration. In particular, they investigated *microfiltration*.⁴ the filtration of molecules in the size range 4,000–6,000 Angstroms, molecules much larger than the 10–150 Angstrom size range for which ultrafiltration was appropriate. When whole blood was filtered by the Amicon process over a microfiltration membrane, blood plasma was successfully separated from whole blood. Blood plasma, which consists of water, a variety of salts and many types of protein, passed through the membrane, while the blood cells (red and white) flowed on. Previous to the application of Amicon's parallel-flow technique, it had been impossible to separate blood plasma from whole blood by a filtration process without rupturing the blood cells, which then contaminated the blood plasma.

On the basis of its discovery Amicon applied for a patent on a new apparatus and process for the separation of blood plasma from whole blood, and in December 1972, the United States Patent Office granted patent number 3,705,100 which protected Amicon's discovery. A copy of the patent is reproduced as **Exhibit 4**. Amicon was subsequently granted patents on the invention in Canada, Japan, France, West Germany, and the United Kingdom.

A key advantage of the Amicon filtration system over existing technology was the continuous nature of the filtration process. The existing method for obtaining blood plasma involved a number of stages. First, whole blood was collected from the patient or donor. Second, the whole blood was placed in a centrifuge and spun at moderate velocity to effect a separation of blood plasma from the blood cells. Third, the blood plasma was removed from the separation vessel, and fourth, in certain applications, the blood cells were returned to the patient or donor. Use of the Amicon filtration system would permit whole blood to be withdrawn from the patient or donor, filtered to remove blood plasma, and the blood cells returned in one continuous operation.

³Of approximately 20 centers which were sustaining a total of 75 patients without normal kidney function worldwide, 85% were located in Europe and 15% in the United States.

⁴Although the production of ultrafiltration membranes was limited to Amicon, Dorr-Oliver and Romicon, since the original patents were still in force, the production of microfiltration membranes was not controlled by patents, and they were available from many suppliers.

Because of a long history of unsuccessful attempts to separate blood plasma from whole blood using microfiltration membranes and conventional filtration processes, suppliers of products to the blood-plasma collection industry and some government funding agencies had expressed skepticism over the practical value of Amicon's process. However, as the result of a study funded by the National Institute of Health (NIH),⁵ researchers at the American National Red Cross had concluded that Amicon's process claims were fully justified. Further, Red Cross technical staff personnel were actively promoting the concept of blood-plasma collection by filtration.

The Red Cross has subcontracted a portion of this study to Amicon and, as a result of Amicon's work, two new patent applications, generally known as improvement inventions, had been filed in early 1978. One, coauthored by Red Cross and Amicon personnel, specified the exact conditions under which a commercial-size plasma separation system would have to operate, while the second, authored solely by Amicon personnel, specified a particular filter design which would allow blood-plasma separation in commercial quantities. Should these two patent applications be granted, however, the patents would be owned by NIH, since it had funded the study which gave rise to them.⁶ For an optimum commercial system to be manufactured and used, however, all three patents (including Amicon's patent number 3,705,100) would likely be required.

Mr. Jacobs said that there were four major applications for which the separation of blood plasma from whole blood might offer considerable benefits; three human applications, blood-plasma collection, blood-plasma analysis, and blood-plasma therapy, and animal blood-plasma collection. Of these applications human blood-plasma collection offered the greatest immediate potential.

Human Blood-Plasma Collection⁷

In 1978 the United States led in the production and distribution of blood-plasma fractions, a series of natural drug products which were obtained by the secondary processing of human blood plasma⁸. Sales of blood-plasma fractions by United States organizations in 1977 were estimated at \$240 million and were 96% of total world volume. Approximately 20% of this amount was exported. United States market leadership had resulted from a number of factors: technological leadership, more liberal laws relating to blood than in some other countries,⁹ and a general inability of other countries to meet local needs because of the high growth in the demand for blood-plasma fractions. As an example, Japan imported over 40% of its plasma needs in 1977. Because of the growth of demand abroad, several United States plasma processors had been acquired by Japanese, German, and French companies.

While the centrifuge process was the sole method by which blood plasma was separated from whole blood, two distinct plasma collection systems existed. In one system whole blood was obtained from *unpaid donors*. Typically, a donor gave one-half liter of blood in a 30-minute session, a maximum of once per month. Donated blood was either distributed to hospitals for blood

⁵A division of the U.S. Department of Health, Education, and Welfare (HEW).

⁶For many years it had been the practice of the United States Government to grant no-fee, nonexclusive licenses to anyone interested in operating within the claims of one of its patents. Thus, patents owned by the government essentially had the status of scientific publications. However, HEW more recently had begun to release patent rights to nonprofit organizations such as the American National Red Cross and to encourage such organizations to collect fees for licenses to use such patents. HEW had not yet decided whether it would release the two improvement inventions on blood separation technology if the patent applications were granted.

⁷Much of the data for this section was obtained from an article in *Business Week*. September 11, 1978.

⁸Any particular blood-plasma fraction consisted of a defined molecular weight range of protein molecules.

⁹For example, the separation of blood plasma from whole blood was banned in Italy; in Japan payment for blood was not allowed and French law forbade profiting from the sale and processing of human blood.

transfusions, or it was sent to contract processors who separated off the blood plasma and then extracted the blood-plasma fractions. From one-half liter of whole blood slightly over 300 milliliters of blood plasma could be obtained. Usually the blood collection agency distributed the blood-plasma fractions along with whole blood for blood transfusions.

Blood collection and distribution were dominated by the American National Red Cross which accounted for over 50% of voluntary blood donations¹⁰. Hospitals and independent blood collection agencies accounted for the remainder. The Red Cross had a network of 57 regional blood centers and almost 4 million donors. Industry observers estimated that over 50% of the whole blood collected by the Red Cross was sent to contract processors for blood-plasma fraction production. The Red Cross was reported to have paid contract processors \$9.1 million for blood-plasma fractions in 1977, which it then sold for \$29.4 million.

In the second system, whole blood was obtained from *paid donors* for the specific purpose of producing blood plasma. As with the former system, one-half liter of blood, the maximum allowable under federal law, was withdrawn from the donor. However, in the case of paid donors, the blood was immediately centrifuged, and the blood cells were reintroduced to the donor. The effect of this procedure was to allow a second half liter of blood, which in turn was immediately centrifuged for blood-cell reintroductions, to be collected in the same session. A donor, who was usually paid between \$7–10, could give blood plasma up to twice per week.

Blood plasma was obtained from donors in 500 to 600 commercial blood-plasma collection centers nationwide, mostly located near universities or in the low-income areas of urban centers. A typical center collected 10,000 liters of blood plasma annually, the range being 3,000 to 20,000 liters. Independent commercial centers sold blood plasma either under long-term contracts for between \$47–50 per liter, or in the spot market for \$50–55 per liter. Approximately 65% of the commercial centers were independent operations, 20% were owned and operated by Maynard-Smith Inc. and 10% by Jackson Industries.

Both Maynard-Smith and Jackson Industries were involved in other aspects of blood collection and processing. Maynard-Smith produced disposable blood collection kits consisting of catheters, tubing, storage vessels, and anticlotting chemicals, most of which were supplied to its own plasma collection centers. It also processed blood plasma into blood-plasma fractions. Maynard-Smith had recently been acquired by Euro Drug, a large European pharmaceutical company.

Jackson Industries, a billion-dollar company, also produced disposable blood collection kits, and was estimated to have a 90% share of the sales of kits to the independent plasma collection centers, the Red Cross, and other centers which collected just whole blood. Jackson Industries was also a major processor of blood plasma into plasma fractions and accounted for 25% of United States fractionating capacity. It was the major customer of the independent plasma collection centers for blood plasma, and the major contract processor for the Red Cross and other voluntary blood collection organizations.

A furor had erupted in the industry in early 1978 with the announcement that Jackson Industries and the American National Red Cross planned to build a one million liter blood-plasma fractionating plant in a \$40-50 million joint venture. Such a plant would account for about 20% of industry fractionating capacity. By controlling the fractionating of its own blood, the Red Cross believed that it could achieve better quality control and reduce the \$18-per-liter fractionating price by \$5 per liter. However, independent companies in the industry had charged that the American

¹⁰Nearly half of the Red Cross/ 1977 revenues of \$350 million was accounted for by blood-related activities. For its nonblood-related activities the Red Cross was the leading beneficiary of United Way funds, receiving a total of \$126 million in 1977.

National Red Cross was trying to drive them out of business, and the joint-venture proposal was under review by the Justice Department at the request of the Red Cross.

The Amicon Process. Mr. Jacobs believed that the Amicon filtration process offered considerable benefits both to the voluntary blood collection agencies and to the commercial plasma collection centers.

The distinctive feature of the Amicon process was its continuous nature. It would permit blood plasma to be collected directly from a donor, while the blood cells were simultaneously reintroduced to the donor. In this manner 625 milliliters of blood plasma could be collected in a 30- to 45-minute session and a donor could give plasma up to twice a week.

To the voluntary agencies, the Amicon process opened up the possibility of direct blood-plasma collection. Such collection had not previously been possible, since the voluntary agencies felt they could not expect a volunteer donor to spend the two hours required by the centrifuge technique. Further, volunteer agencies presently obtained little more than 300 milliliters of blood plasma per sessions from the allowable half liter whole blood donation. Use of the Amicon system would make it possible to collect over 600 milliliters of blood plasma in a single session. Finally, blood-plasma collection potential would be significantly increased since donors could provide blood plasma more frequently than in the past.

To the commercial plasma collection center, the most important advantage of the continuous system was the complete elimination of any possibility of the donor receiving someone else's blood cells. Although there were very careful checks in the present system, there was always the possibility of a slip-up which could have disastrous consequences for the donor. Second, the time saving inherent in the continuous process would significantly increase the capacity of a blood collection center and greatly reduce donor inconvenience.

On the other hand, the cost to the collection centers of disposable items would increase from \$12-20 per application¹¹. The difference would be partially offset by a \$2 to \$3 labor cost savings. In addition, centers would also have to purchase electronic control systems (one per bed) to monitor blood flow rates. Jacobs thought, however, that the cost of these systems would be equivalent to that required for centrifuges which would no longer be required.

The overall attractiveness of increased safety and shorter plasma collection times, versus higher total costs for disposables, was subject to much debate. Some industry representatives had told Jacobs that plasma collection centers would willingly pay the higher cost to eliminate entirely the risk of returning the wrong cells to a donor and to attract more donors by shortening the time required for donation. Others said that the low profit margins of the collection centers would severely restrict the number of centers which could pay the added cost of the new filtration system.

Mr. Jacobs estimated, however, that if all the commercial collection centers in the United States adopted the Amicon process there would be an annual requirement for 4 million filtration units (the microfiltration membranes embedded in plastic casings). Further, should the American National Red Cross and the other voluntary organizations switch a significant proportion of their whole-blood collection programs to blood-plasma collection, there would be a considerably increased demand there, as well, for filtration units. However, before the Amicon process could be used commercially in the United States, FDA approval would have to be obtained. No such impediment existed for sale to overseas markets, provided that the filtration units were manufactured abroad. Jacobs believed that demand from overseas could be substantial.

¹¹The \$8 increase was a net figure. A disposable filtration device, consisting of a specially designed microfiltration unit embedded in a plastic casing would be required. However, certain of the other disposable items (catheters, tubing, storage vessels, anticlotting chemicals) would no longer be needed.

Mr. Jacobs' belief that considerable sales potential existed had been confirmed in a recent confidential study issued by a California-based investment company. The author had estimated that in 1983, 1.8 million filtration units could be sold worldwide, approximately 50% in the United States and 50% in Europe.

Human Blood-Plasma Analysis And Therapy

Two additional human applications for the Amicon process were blood-plasma analysis and blood-plasma therapy. Many blood analyses were more efficiently carried out on blood plasma than on whole blood. Under the present system, samples of blood were drawn from a patient in a hospital or doctor's office and subsequently centrifuged in a laboratory to obtain blood plasma. Jacobs believed that the small quantities of plasma required for analytic purposes could be obtained more simply, using the Amicon filtration technique.

Although he was unsure of the potential demand for filtration units that would be involved in the analytic market, Jacobs thought that it might be substantial because of the advantages of the continuous system. This was especially true for automated blood analyses, where the centrifuge stage represented the only discontinuous step in an otherwise continuous operation. He believed that hospitals represented the most likely customer group, although he did not rule out the possibility of sales to physicians, particularly group practices or clinics where patient throughput was at a high level. FDA approval would also be required for this application, but diagnostic devices required significantly less testing than medical devices.

Many blood disorders required the hospital treatment or replacement of blood plasma. The current therapeutic process called for collection of whole blood and separation of blood plasma from blood cells by centrifuge. Then, either the patient's plasma was treated by exposure to enzymes or other active chemicals, recombined with the blood cells, and reintroduced to the patient, or the patient's own plasma was discarded and his or her blood cells combined with donor-supplied plasma and then reintroduced to the body. The Amicon separation technique would permit the whole process, regardless of whether the plasma was treated or discarded, to be performed continuously, and would considerably reduce both the treatment time and patient trauma.

Amicon was currently working with one doctor who was very excited about the potential for this process. However, Jacobs said that commercial development was many years away since much more research was necessary and, again, FDA approval would be required. He was unable to put a figure on the potential market size but thought that it, as well as the analytic market, might be considerable. He anticipated that hospitals would represent the major customer group.

Animal Blood-Plasma Collection

A final application for Amicon's blood-plasma separation process had recently been identified by Dr. Solomon, Amicon's director of research. Dr. Solomon said that an industry had developed for the production of antibodies for use in medical research and in numerous diagnostic tests. These antibodies were grown in specially-kept sheep, pigs, and goats which were housed in farms devoted entirely to antibody production. The antibodies grew in the animal's blood plasma and periodically the animals were bled and the blood plasma obtained by a centrifuge technique, similar in concept to that used for human donors. It appeared that Amicon's process, which allowed for continuous plasma collection and blood-cell reintroduction, could offer considerable benefits to farm owners. Although a prototype product for animal blood-plasma separation had not yet been developed, Dr. Solomon said that he was being besieged by telephone calls from farm owners willing to pay very high prices for a working system. An attraction of this application to Amicon was that in contrast to the human blood-plasma collection application, there was a possibility that filtration devices could be sold in the United States without FDA approval.

Alternative Courses Of Action For Amicon

With each of the four fields of use presenting important opportunities for Amicon, Jacobs had to decide which seemed most promising. Then it would be essential to make some difficult choices regarding strategy. Direct entry was one approach he would consider; licensing and joint ventures were other possibilities. Even within these options there were further choices to be made.

Human Blood-Plasma Collection

Jacobs said that the large potential market for human blood plasma, together with the anticipated growth, made *direct entry*, which would include manufacturing and selling the filtration units, a tempting proposition. Since Amicon would be protected by its patents for another 11 years it could, by 1989, have built up a dominant position. On the other hand, he was concerned about the resource commitment which direct entry would entail. Amicon would have to perform the engineering development work necessary to turn its laboratory prototype filtration units into full-scale commercial products. Second, it would have to set up a production line to manufacture high volume, low cost plastic parts.¹² Jacobs said that Amicon could probably purchase the microfiltration membranes from an outside supplier but would have to set up two assembly operations: one to construct filtration units from the membranes and plastic parts, and a second to connect the filtration units to the auxiliary plastic tubing sets to produce blood-plasma collection kits. The tubing and other components would be purchased from outside suppliers. A clean-room manufacturing facility would have to be constructed to create a dust- and bacteria-free environment. A sterilization system would also be required. He estimated the total capital cost at \$1–2 million, although a small-scale operation might be put in place for only \$500,000. Finally, Amicon would have to develop a sales force to visit blood collection centers, and a service operation would be needed to solve in-use problems and to ensure that plasma separation was being properly carried out. In particular, a service facility would be required to assure prompt maintenance, repair and/or replacement of control systems components. The cost to Amicon of having a sales or service person in the field was approximately \$40,000 per annum, \$20,000 for compensation and an additional \$20,000 for travel and expenses. Jacobs estimated that at a minimum five salespersons, a sales manager and three service personnel would be required. If Amicon did enter directly, Jacobs estimated that it could obtain sales revenue of \$20 for a blood-plasma collection kit, while the direct manufacturing cost would be approximately \$10.

Jacobs was unsure how Maynard-Smith and Jackson would react to Amicon's entry. Since Maynard-Smith produced all of the tubing sets and collection vessels for its own blood collection centers, it might be unwilling to purchase the completed kits from Amicon. Further, in dealing with the independent centers and the Red Cross, Amicon would be competing head-to-head with Jackson which currently supplied disposable kits to the major part of the market. An even more critical consideration was that Amicon's direct entry into the plasma collection market would almost certainly stimulate efforts by Maynard-Smith and Jackson to "invent around" the Amicon patent and develop their own filtration devices.

A second alternative which Mr. Jacobs was considering was a *joint venture*, but he was unsure whether Amicon's patent by itself was sufficient for a joint-venture partner to be interested. Further, there was the problem of whom to approach to be a joint-venture partner. Should it be Maynard-Smith, Jackson, or some third party not currently directly involved with blood-plasma production? Indeed, since the Red Cross was planning to enter the plasma fractionating business, perhaps it would be interested in becoming a partner in the production of filtration units.

¹²Plastic materials used for this purpose would have to meet medical grade specifications. These materials were particularly difficult to mold.

Finally, Mr. Jacobs was considering the possibility of *licensing*. The major advantage of this approach was that Amicon could receive royalty payments with little additional investment of money or people. However, in 1989, all royalty payments would cease. Further, Amicon would no longer be at the cutting edge of blood-plasma separation technology.

While Jacobs found some aspects of licensing attractive, this option presented a number of serious problems. The most serious problem revolved around the nature of the rights to be granted by the licensor, Amicon, to a licensee. In particular there was the question of "field of use." While Jacobs might be interested in making license agreements in the human blood-plasma collection application, he might wish to exclude any licensee or licensees in this application from the analytic, therapeutic and animal blood-plasma collection applications. While this might seem to be simple enough in principle, in practice it was very difficult to separate fields of use. For instance, if a hospital purchased a filtration device for blood-plasma collection, there was nothing to stop it from using the device for therapeutic purposes, if the device was suitable. Jacobs thought it might also be very difficult to draft language which would clearly delineate applications.

The agreements would also have to avoid carefully the implication that they represented a "carving up" of markets among competitors, which would be a violation of antitrust law. Although the Justice Department had recently acknowledged informally that field-of-use licensing was not automatically an antitrust violation, certain agreements had been challenged on such grounds.

Furthermore, if another competitor produces and sold devices covered by the patent and Amicon filed suit to stop such activity, the infringer would be certain to examine carefully any license agreement, hoping to find a possible antitrust violation which would prevent Amicon from enforcing the patent. There was, however, a possibility that the filtration devices designed for different applications would be sufficiently unique so that product specifications could be written into a license agreement, thus, in effect, delineating fields of use.

A second problem that Jacobs would face if he went the license route was the question of what sort of license to grant. There were four types: exclusive, exclusive but for Amicon, limited exclusive, and nonexclusive. If Amicon granted an exclusive license, just one licensee would enjoy all of the rights under the patent that currently accrued to Amicon. If a license which was exclusive but for Amicon was granted, just the one licensee and Amicon would be the only companies able to enjoy the patent rights. If limited exclusive licenses were granted, Amicon would specify how many companies could enjoy the patent rights at the time the first company signed a license agreement. It could not subsequently increase the number of licensees beyond that stipulated number. Finally, if nonexclusive licenses were granted, Amicon could offer licenses to whomever it chose. If Amicon entered into a nonexclusive license arrangement, the value of the agreement to a licensee would, of course, be lower and the price which Amicon could command would be less than if an exclusive license was granted. Further, Mr. Jacobs was not certain whether the protection of exclusivity was required to encourage a potentially strong licensee to make the investment necessary to commercialize the product. On the other hand, by granting more than one license, Amicon might encourage competition and thus hasten market development.

If a license or licenses were to be granted, Jacobs had to decide how to charge for the license and how much to charge. As a rule of thumb he believed that licensors typically received 10–35% of the profits that the licensee made during the life of the license. However, if the license fee was set at some percentage of the profit, Jacobs was concerned that there would be serious accounting problems in determining the true "profit" from which the royalty would be computed. If Maynard-Smith was a licensee, the problem was doubly complicated since the price at which the filtration units were sold would be an administered transfer price from one division to another and not a market-determined price. Finally, to the extent that any licensee sold the filtration unit as one part of a blood-plasma collection kit, there would be a problem in determining the value of the filtration unit.

Regardless of the basis on which the license fee was determined, there was the question of how high the fee should be. A high fee might bring in a good return of royalties to Amicon, but too high a fee might discourage companies from taking up a license and might lead them to attempt to develop alternative filtration devices outside the patent. Further, Mr. Jacobs was concerned that any licensee would, in fact, commit sufficient resources to bring products to market under the terms of the license. Therefore, he had to find some way of ensuring that any licensee would move quickly to gain FDA approval and place products on the market so that Amicon could start to receive earned royalty payments. Jacobs thought that a minimum royalty payment might be the way to solve this problem. If he did institute a minimum royalty payment, Mr. Jacobs had to decide how much it would be, and whether it should be constant or based on an ascending or descending scale.

Mr. Jacobs also had to take into account the possibility of patent infringement or circumvention. Under patent law, if the optimal system involved operation within the patent, and the process was used just outside the patent claims in a clear attempt to get around the patent, then it would be deemed an infringement. However, the possibility of a legal infringement was reduced the further outside the claims that the process was operated. While Amicon's scientists had assured Mr. Jacobs that the patent was "watertight" and that operation outside the patent was impossible, he was nevertheless concerned about this issue after learning that one or more companies were already conducting tests with exactly that objective in mind.

In light of all of the considerations involved in granting a license or licenses Jacobs had to decide whom to license. The major candidates for a license were Jackson, Maynard-Smith, and a German company, Behrstein. While Mr. Jacobs believed that an exclusive license, since it provided a monopoly for the licensee, should be attractive, neither Behrstein nor Maynard-Smith had shown a strong interest in an exclusive license. His early discussions with Jackson, however, led Mr. Jacobs to believe that it would be interested in such an arrangement.

The major advantage of making an agreement with Jackson was its dominance in the market. It was closely tied to the independent commercial blood collection centers and to the Red Cross. If it decided to push the Amicon filtration process, Jacobs had little doubt that impressive sales would be forthcoming, although it would take significant sales and service efforts from Jackson to persuade the independent centers to adopt the invention. If Maynard-Smith took up the license, the adoption of the invention might be extremely fast since Maynard-Smith owned its own blood collection centers. However, it accounted for only 20% of the blood plasma collected directly on an individual basis from paid donors. On the other hand, for years Maynard-Smith had tried to lessen Jackson's hold on the independent centers with very little success, and the new filtration process might enable it to do so.

Behrstein was a major company in the supply of blood collection kits to blood-plasma collection centers in Germany and saw a license under the Amicon patent as a way of gaining a foothold in the blood-plasma business in the United States, as well. However, Mr. Jacobs thought it unlikely that Behrstein would achieve significant penetration in the next few years, even if it had exclusive rights under the Amicon patent. Its strength in Germany, however, raised the question in Mr. Jacobs' mind as to whether he should put territorial restrictions on licenses. He knew that although Jackson and Maynard-Smith were active in the blood-plasma business abroad, other locally based companies might be significantly stronger. He guessed that the potential for Amicon's process in blood-plasma collection overseas might be considerable.

Human Blood-Plasma Analysis And Therapy

Mr. Jacobs said that he had had extended discussions with representatives of a small Boston company about granting a license to exploit the analytic market. This potential licensee was anxious to sign a patent agreement, and Jacobs knew from experience that an enthusiastic potential licensee could easily lose interest if negotiations were long and drawn out. Therefore, he knew that he had to

make a decision quickly. He was concerned, however, about the implications of granting a license in the analytic market for possible licensing in the blood-plasma collection market. His concern was about the division of exclusive rights in a patent for it was as yet unclear whether different products would be used in the two applications. Further, since the customers for blood-plasma analysis devices were largely concentrated in the hospitals on which Amicon salespeople now called, he wondered if perhaps Amicon should undertake to develop this market itself.

Commercial development of the blood-plasma therapy market was some years away, but Jacobs wanted to retain the rights in this field for Amicon. He saw a good fit between this application and the blood filtration application for artificial kidneys which the company was developing. Further he anticipated that NIH would be willing to fund research and clinical testing in conjunction with outside medical researchers as it had done in the kidney field.

Jacobs realized, however, that he should decide now on how this market should be approached if the technical development efforts were successful. Otherwise he might foreclose his options with decisions made regarding other applications. He thought that Amicon might wish to enter the market directly although a joint venture might also be a possibility. He had already been approached by some large drug companies about the possibility of entering into joint-venture arrangements.

Finally, there was the possibility of licensing, but then there was the question of whom to license. He considered Jackson as one possibility but thought that one complicating factor regarding other potential licensees was that Jackson's management, apparently aware of the one doctor's research efforts, might wish to include access to the therapeutic application in any license which it negotiated.

Animal Blood-Plasma Collection

The identification of the animal application was so recent that Amicon executives had given little thought on how to proceed. They were fairly confident, however, that they could successfully insulate this application from the human applications and that providing for field-of-use restrictions would not be difficult, whichever commercialization route they chose.

Exhibit 1

	<u>Income Statements</u> (thousands)			
	<u>1977</u>	<u>1976</u>	<u>1975</u>	<u>1974</u>
Sales	\$11,232	\$ 8,873	\$ 7,435	\$ 7,141
Cost of Goods Sold	4,891	3,688	3,138	3,021
G. S. and A.	4,621	3,651	3,085	2,841
R. and D. Expense	647	592	574	570
Interest Expense	115	107	113	094
Currency Loss (Gain)	<u>33</u>	<u>42</u>	<u>19</u>	<u>13</u>
Net Income Before Taxes	925	877	506	628
Provision for Taxation	<u>352</u>	<u>395</u>	<u>222</u>	<u>234</u>
Net Income	\$ 573	\$ 482	\$ 284	\$ 393

Exhibit 2

	<u>Balance Sheet</u> (thousands)			
	<i>1977</i>	<i>1976</i>	<i>1975</i>	<i>1974</i>
CURRENT ASSETS				
Cash	\$ 454	\$ 413	\$ 362	\$ 77
Accounts Receivable	1,949	1,440	1,335	1,185
Inventories	1,702	1,533	1,263	1,360
Prepaid Expenses & Deposits	229	163	155	118
Deferred Taxes	—	—	90	—
Net Assets from Discontinued Operations	<u>—</u>	<u>—</u>	<u>23</u>	<u>208</u>
Total Current Assets	\$4,334	\$3,549	\$3,228	\$2,949
FIXED ASSETS				
Land	\$ 161	\$ 161	\$ 142	\$ 142
Buildings, Machinery & Fixtures	<u>2,976</u>	<u>2,476</u>	<u>2,286</u>	<u>1,128</u>
	3,137	2,637	2,428	2,270
Less Accumulated Depreciation	(1,259)	(1,054)	(869)	(686)
	1,878	1,583	1,558	1,584
Investment in Affiliates	511	540	360	333
Deferred Pre-Operating Costs	<u>23</u>	<u>—</u>	<u>—</u>	<u>—</u>
	\$6,476	\$5,672	\$5,146	\$4,876
CURRENT LIABILITIES				
Notes Payable and Current Installment on Long-Term Debt	\$ 119	\$ 38	\$ 31	\$ 996
Accounts Payable	692	441	366	480
Income Taxes	235	137	73	41
Accrued Expenses & Other Liabilities	<u>1,637</u>	<u>1,084</u>	<u>0,431</u>	<u>1,834</u>
LONG-TERM DEBT	\$1,652	\$1,606	\$1,687	\$ 620
	<u>—</u>	<u>—</u>	<u>—</u>	<u>—</u>
STOCKHOLDERS' EQUITY				
Common Stock Outstanding	\$ 188	\$ 187	\$ 187	\$ 187
Additional Paid-in Capital	1,102	1,088	1,088	1,085
Retained Earnings	<u>2,128</u>	<u>1,611</u>	<u>1,157</u>	<u>1,036</u>
Total Stockholders' Equity	<u>3,418</u>	<u>2,886</u>	<u>2,433</u>	<u>2,308</u>
	<u>\$6,746</u>	<u>\$5,672</u>	<u>\$5,146</u>	<u>\$4,876</u>

Exhibit 3 Percent of Corporate Sales and Operating Profits by Business Segment, 1973–1977

	<u>1976</u>		<u>1975</u>		<u>1974</u>		<u>1973</u>	
	<u>Sales</u>	<u>Operating Profit</u>	<u>Sales</u>	<u>Operating Profit</u>	<u>Sales</u>	<u>Operating Profit</u>	<u>Sales</u>	<u>Operating Profit</u>
Scientific Systems Division	59%	64%	61%	66%	62%	73%	58%	70%
Polymer Products Division	38%	39%	36%	33%	31%	17%	35%	20%
Research Services & Royalty Income	3%		3%	1%				
1977		(3%)			7%	10%	7%	10%
							8%	

Exhibit 4

Dec. 5, 1972

W. F. BLATT ET AL
BLOOD FRACTIONATING PROCESS AND APPARATUS
FOR CARRYING OUT SAME

3,705,100

Filed Aug. 25, 1970

2 Sheets-Sheet 1

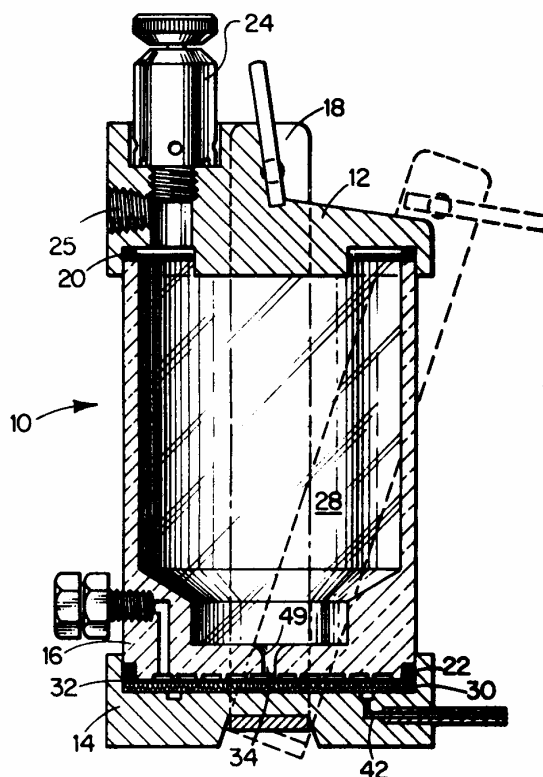


Fig. 1.

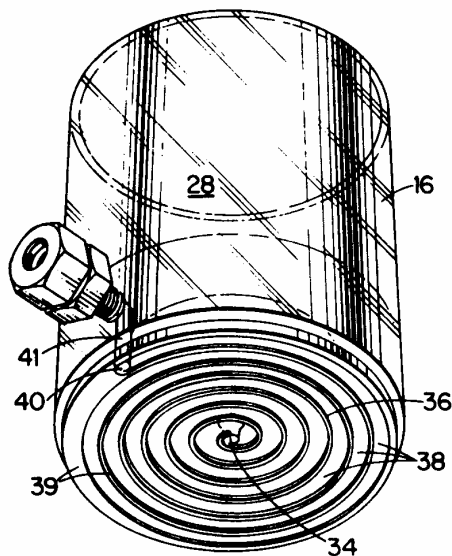


Fig. 2.

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Exhibit 4 (continued)

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W. F. BLATT ET AL
BLOOD FRACTIONATING PROCESS AND APPARATUS
FOR CARRYING OUT SAME

3,705,100

Filed Aug. 25, 1970

2 Sheets-Sheet 2

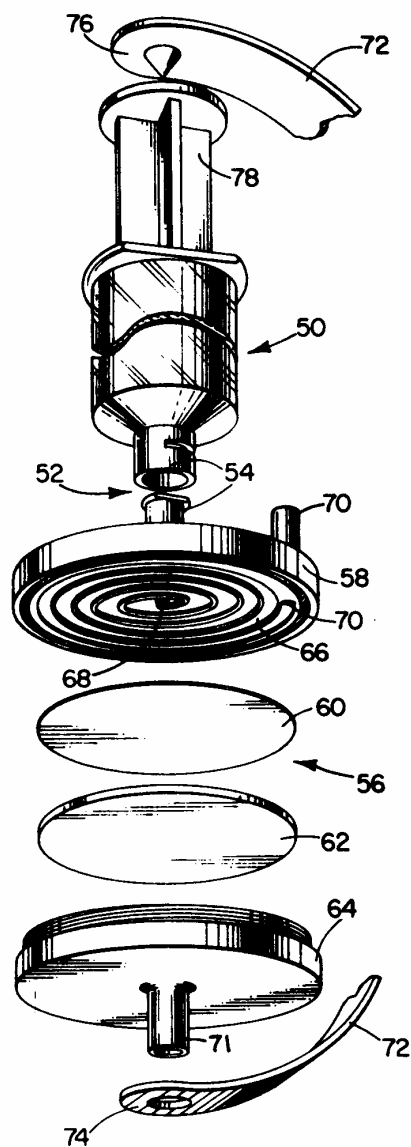


Fig. 3.

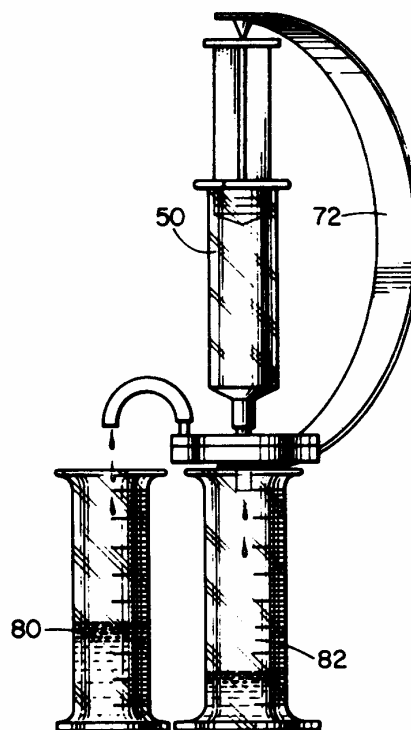


Fig. 4.

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Exhibit 4 (continued)

United States Patent Office

3,705,100

Patented Dec. 5, 1972

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3,705,100

BLOOD FRACTIONATING PROCESS AND APPARATUS FOR CARRYING OUT SAME

William F. Blatt, Framingham, Edward A. Agranat, Weston, and Peter N. Rigopoulos, Melrose, Mass., assignors to Amicon Corporation, Lexington, Mass.

Continuation-in-part of applications Ser. No. 828,935, May 29, 1969, and Ser. No. 833,090, June 13, 1969. This application Aug. 25, 1970, Ser. No. 66,675

Int. Cl. B01d 13/00

U.S. Cl. 210—23

8 Claims

ABSTRACT OF THE DISCLOSURE

A process for separating blood plasma from whole blood that dispenses with the known centrifugal-separation techniques and involves passing whole blood along a flow path which is shallow and substantially parallel to the upstream side of filtration membrane, recovering plasma from the downstream side of said membrane, and recovering the retained blood components (formed elements) from the upstream side of the membrane.

This application is a continuation-in-part of U.S. Ser. No. 828,935, filed May 29, 1969, now abandoned, and of Ser. No. 833,090, filed June 13, 1969, now abandoned.

When obtaining blood from a blood donor, it is very often desirable to be able to return the cellular components to the donor so that more frequent bleedings can be made. When only the plasma component of the blood is desired for emergency use, the formed elements of the blood (which include the red blood cells, white blood cells and platelets) can be discarded or used for other purposes or can profitably be returned to the donor. Such a return is particularly important because (1) it allows the donor to recuperate to a state where he can donate again within two weeks rather than in about 2 months as is the case when the non-plasma component of the blood is not returned to him, and (2) it avoids the temporary weakness suffered by some donors after they donate a pint of blood. The importance of a donor's being able to contribute blood at relatively frequent intervals is obvious in circumstances such as those wherein injuries are incurred during military operations or wherein a donor bears a rare blood-type for which an emergency need exists.

However, blood fractionating of the type described is not used as frequently as desirable because no really convenient means for carrying out the process has been available. In general, this type of blood-fractionating has been done by

- (1) transferring the blood from a donor into a blood bag by means known to most blood donors, then
- (2) transferring the blood bag into a centrifugal separating apparatus, then
- (3) "spinning" the blood at a rate which optimizes the separation of plasma from other blood components, but substantially avoids damage to blood cells, then
- (4) separation of plasma by bag compression or withdrawal to a receiving vessel, and finally
- (5) returning the formed elements back into the patient by the usual transfusion techniques.

Not only does this process involve relatively expensive apparatus, but it also comprises a sufficiently large number of handling steps to significantly increase the chance of contamination and/or cellular damage in the relatively crude environments of the type that may be encountered at accident scenes, in military operations, etc.

Moreover, there are many situations in which it is desirable to separate blood components without returning any of them to the donor in order to use diagnostic tests

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without interference from either the formed cell or plasma components thereof.

The present invention provides a process and apparatus for simple fractionation of whole blood into a plasma component and a cellular component while subjecting the components to only very slight stress. The present invention is furthermore readily applicable to blood-donation procedures, making it possible to return the non-plasma component or fraction to the donor virtually simultaneously with the donation.

The process of the present invention comprises conducting the whole blood in a flow path parallel to one face of a porous filter membrane having effective pore diameters from 0.1 to 0.8 micron, the path having a maximum depth of 20 mils measured vertically from the face of the membrane, collecting from the opposite face of the membrane the plasma component, and collecting from the end of said flow path the cellular component while maintaining the pressure differential between opposite faces of the membrane from 1 to 15 p.s.i. For best results the rate of flow of the whole blood across the face of the membrane is maintained from 2 to 50 ft. per minute and the pore diameter is from 0.4 to 0.7 micron. The precise diameter of the pores within the stated ranges of size which gives best results depends upon the precise pressure differential employed, higher pressure differentials within the stated range requiring smaller pore diameters. The pressure differential is critical because it provides the driving force for controlling the velocity of the blood across, and plasma through the membrane, and also affects the degree of hemolysis which occurs during filtration.

It is essential that the blood being filtered travel in a path substantially parallel to and within 20 mils of the membrane surface. Attempts to utilize the same membranes under conditions whereby the whole blood is forced through the membrane by conventional filtration techniques (i.e., putting the blood in a reservoir over filtration membrane and applying a pressure difference across the membrane to push or pull the plasma fraction through the membrane) results in almost immediate "plugging" of the membrane.

The term "filtration membrane" is used in this application to means that class of filters normally supplied in thin sheet form and capable of effecting separation of very small particulate or molecular components from suspensions or solutions. Both anisotropic and depth-filter membranes are included within this description. The former type of membrane is preferred when conveniently available, but a particularly surprising feature of the invention is that homogeneous depth filters may be utilized in the blood separation process.

The filtration process of the invention is carried out at relatively low pressure differentials, e.g., from 1 to 15 p.s.i., as measured both from one side of the filter membrane to the other and as measured from the inlet of the whole blood passage to the outlet for the blood fraction which fails to pass through the filter membrane. As a matter of convenience, both the receptacle for the filtrate (plasma) and for the rejected blood (non-plasma fraction) are preferably maintained at atmospheric pressure. Pressure differentials near the lower end of this range, i.e., from 1.5 to 5 p.s.i. are most advantageous, in part because they can be utilized in equipment which is less rigorously designed to avoid undue stress on the cells contained in the blood being fractionated. Likewise the velocity across the face of the membrane is relatively low, i.e., in the range of from 2 to 50 feet per minute. Under these conditions, the flow is substantially laminar. In the more preferable embodiments of the invention the blood, after passing over the surface of the membrane, is recycled back to the whole blood reservoir. The velocity of the stream being dis-

Exhibit 4 (continued)

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sipated in the contents of the reservoir aids in keeping the blood mixed well.

In order to point out more fully the nature of the present invention, the following specific example is given as an illustrative embodiment of the present process and products produced thereby.

FIG. 1 shows a view in elevation, partly in section, of a thin channel ultrafiltration cell useful for carrying out the process of the invention.

FIG. 2 is a perspective view from the bottom of the reservoir and flow-directing means showing the apparatus of FIG. 1.

FIG. 3 is an exploded view in perspective showing a novel apparatus useful in the process of another embodiment of the invention which comprises a means to attach a hypodermic needle thereto.

FIG. 4 is a view in elevation showing the apparatus of FIG. 3 in operation.

Referring to FIGS. 1 and 2, it is seen that an ultrafiltration cell 10 comprises a top cap 12, a bottom cap 14 and a cylinder assembly 16. The cylinder is compressed and sealed between caps 12 and 14 by means of toggle clamping assembly 18, top O-ring seal 20, and bottom O-ring seal 22.

Top cap 12 comprises a pressure relief valve 24 and a means to drive fluid across the membrane comprising a port 25 adapted for connection to a pressurized gas source for pressurizing liquid in reservoir 28.

Resting on bottom cap 14 is a macroporous support plate 30 formed of sintered polypropylene. Over plate 30 is a cellulosic ester filtration membrane 32 having a mean pore size of 0.45 ± 0.02 micron and available from Millipore Corporation under the trade designation HAW PO 9025. Lower O-ring seal 22 is compressed against the outer periphery of membrane 32, thereby providing an efficient edge sealing means.

Cylinder assembly 16 comprises reservoir 28 and an aperture 34 leading from reservoir 28 into a spiral flow path 36 which is formed by spiral grooves 38 on the bottom surface 39 of assembly 16. This flow path 36 is 0.125 inch wide and 0.010 inch (10 mils) high. It follows a spiral path in a plane parallel to the membrane surface, terminating at a fluid outlet port 40 through which the retained liquid may, via conduit 41, be collected or recycled for another concentrating step. Filtrate, i.e., that fraction of material which comes through the filter is carried out of the cell through conduit 42 which is machined into bottom cap 14.

A sample of whole blood (treated with ACD) was inserted into reservoir 28 and, under a 2 p.s.i.g. driving force, was divided into a plasma fraction and a cellular fraction. The whole blood was forced through aperture 34 in cylinder assembly 16, and thereupon is caused to follow spiral flow path 36 over the surface of membrane 32. The blood plasma fraction passed through the filtration membrane, and was collected through conduit 41 at atmospheric pressure. About 60% of the plasma content of the blood was recovered and there was no evidence of hemolysis in the plasma so collected.

Although the optimum operation of the illustrated device was realized with an operating pressure of from 2 to 4 p.s.i.g., it is stressed that higher operating pressures may be used when particular care is taken to smooth blood-contacting surfaces in such a way as to avoid excessive mechanical shear on the formed elements of the blood. For example, a stream-lined or smooth-surfaced wall 49 with gently rounded corners of aperture 34 is advantageous in this respect. In general, however, a low-pressure process is most desirable for use in emergency blood-donation procedures.

Another embodiment of the apparatus is disclosed in FIG. 3. In this apparatus, which is most useful in analytical work, a hypodermic syringe 50 has been utilized to withdraw a blood sample from a patient. The needle (not

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shown) of the syringe is then removed and the syringe is attached, by means of a fastening means 52, such as Luer lock 54, to filtration cell 56. Filtration cell 56 comprises a top retaining plate 58, filtration membrane 60, a sintered porous polyethylene support disk 62, and a bottom retaining plate 64.

Retaining plate 58 comprises a spiral ridge forming a shallow flow path 66 having a depth of 6 mils, a width of 0.5 cm. and a length of 70 cm. between inlet port 68 and outlet port 70. Retaining plate 64 comprises a filtration outlet port 71.

In order to achieve the most reproducible filtration results, it has been found more desirable to provide the above-described apparatus with a positive pressure control means rather than to rely upon the manual pressure exerted by a number of different operators. Therefore, a spring means 72 is mounted, at one end 74 thereof, on projecting outlet port 70. The other end 76 of the spring is adapted to press on plunger 78 of the hypodermic syringe 50. When spring means 72 is so mounted as to rest on plunger 78, a controlled amount of pressure, about 2.5 p.s.i., is generated for filtering the blood. Another advantage is that one operator can utilize a number of these devices at a single time since they do not require close attention during the filtration operation.

FIG. 4 shows a schematic diagram showing the analytical device of FIG. 3 in operation. A plasma fraction of the blood is being collected in vessel 82 while the other blood components are being collected in vessel 80.

Using the cellulosic ester membrane described above, less than 0.1% hemolysis was observed, and the plasma obtained was not detectably different from that obtained by conventional centrifugation. From a 10 ml. sample of fresh blood of normal hematocrit, there was obtained, in a filtering time of 15 to 20 minutes, approximately 3.0 to 3.4 ml. of plasma. Similar results were obtained using under the same conditions a polycarbonate membrane 1-10 mils thick) having a pore size of 0.5 ± 0.06 microns available under the trademark Nuclepore from the General Electric Company.

Various other advantages and modifications will be apparent to those skilled in the art and fall within the scope of the appended claims.

We claim:

1. Apparatus constructed and arranged to carry out a separation of whole blood into a plasma fraction and a cellular fraction, said apparatus comprising

(1) a reservoir for holding whole blood which is to be fractionated,

(2) a filtration membrane having a pore size from about 0.1 to 0.8 micron diameter,

(3) a flow directing means adjacent one side of said membrane for conducting whole blood from said reservoir across the face of said membrane in a zone having a maximum depth of 20 mils measured vertically from the face of said membrane, and

(4) pressure-generating means constructed and arranged to drive said whole blood to be fractionated through said flow path only within the range of a pressure differential from 1 to 15 p.s.i. and at a flow velocity across the face of the membrane from 2 to 50 feet per minute.

2. Apparatus as defined in claim 1 wherein said reservoir is formed of the barrel of a hypodermic syringe, wherein said pressure-generating means comprises the piston of a hypodermic syringe, and wherein said syringe is detachably connected to said membrane and flow directing means.

3. Apparatus as defined in claim 1 wherein said filtration membrane has a pore size from about 0.4 to about 0.7 micron in diameter.

4. Apparatus as defined in claim 2 comprising additionally a spring for automatically operating the piston of said hypodermic syringe.

Exhibit 4 (continued)

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5. A process for separating blood plasma from the other components of blood comprising the steps of

- (1) conducting whole blood in a flow path which is substantially parallel to the upstream side of a filtration membrane and has a maximum depth of 20 mils measured vertically from the face of the membrane, said membrane having a pore size from about 0.1 to about 0.8 micron in diameter,
- (2) applying sufficient pressure to said whole blood to cause pressure differential from 1 to 15 p.s.i. between upstream and downstream sides of said membrane and to provide a flow velocity across the face of the membrane from 2 to 50 feet per minute,
- (3) recovering plasma ultrafiltrate from the downstream side of said membrane, and
- (4) recovering the retained blood components from the upstream side of the membrane.

6. A process as defined in claim 5 wherein said filtration membrane has a pore size from about 0.4 to about 0.7 micron in diameter.

7. A process as defined in claim 6 wherein the pressure differential is from 2 to 5 p.s.i.

8. In a process for removing whole blood from a blood donor and returning the blood constituents to the donor while keeping the plasma for medical use, the improvement consisting of

- (1) transferring said whole blood from said donor into contact with the upstream side of filtration membrane having a pore size from about 0.1 to about 0.8 micron in diameter,
- (2) conducting said whole blood across the surface of the membrane in a path which is substantially parallel to the upstream side of said membrane and has

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a maximum depth of 20 mils measured vertically from the face of the membrane,

- (3) applying sufficient pressure to said whole blood to provide a pressure differential of 1 to 15 p.s.i. between upstream and downstream sides of said membrane and to provide a flow velocity across the face of the membrane from 2 to 50 feet per minute, and
- (4) recovering the retained blood components from the upstream side of the membrane and transferring them back to the blood donor.

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FRANK A. SPEAR, Jr., Primary Examiner

U.S. Cl. X.R.

210—321, 433, 456

The Licensing Game: *Amicon-Jackson-Hemaglobal*
Supplementary Information

The “Amicon (A)” case ends with Mr. Norman Jacobs, the President, wondering which of the following three strategies *Amicon Corporation* should follow to commercialize its patented blood-filtration technology:

- producing and marketing the products itself;
- collaborating with a joint-venture partner to produce and market the products;
- licensing the technology to another company or companies.

After further thought, Mr. Jacobs and the Board of *Amicon Corporation* decided that it would be best to license out the technology. *Amicon Corporation* then developed a licensing proposal, containing the following terms:

Type of license:	Exclusive
Initial payment:	\$60,000*

Minimum Royalty Payments:

Year 1	\$0
Year 2	\$50,000
Year 3	\$50,000
Year 4	\$120,000
Year 5	\$180,000
Year 6	\$240,000
Year 7 and subsequent years:	\$240,000

Earned Royalty: 8% of sales, adjusted to 5% with increased volume.

Amicon’s terms stipulated a \$60,000 initial payment, plus annual minimum payments from the second year until expiration of the patent. Payments in years two and three, or until FDA marketing approval was obtained, whichever was earlier, would be \$50,000 per year. In the year in which FDA marketing approval was obtained, or Year four, whichever was earlier, the minimum royalty payment would be \$120,000. The next year's minimum royalty payment would be \$180,000 and in each subsequent year until the end of the license, the minimum payment would be \$240,000 per annum.

Earned royalty payments of 8% of sales would be payable on the first \$1,000,000 sales per year, 7% from \$1,000,001 to \$2,000,000, 6% from \$2,000,001 to \$3,000,000 and 5% over \$3,000,000. Earned royalties in any year would be set against the minimum royalties for that year. The license could be terminated by the licensee upon 90-days' notice.

*[*Note: All financial figures in this case are expressed in 1978 dollars. \$60,000 in 1978 would now be worth significantly more than a quarter of a million US dollars ... about \$300,000.]*

This licensing proposal was presented separately to each of *Jackson*, *Behrstein* and *Maynard-Smith*. None of those three companies was willing to enter in to a license agreement with *Amicon* on those stipulated terms. Shortly after receiving their separate licensing offers from *Amicon* (which were both rejected), *Behrstein* and *Maynard-Smith* merged in to a single new company, *Hemaglobal*. *Amicon* then decided to try again, and then entered in to a fresh round of license negotiations with *Jackson* and the newly formed *Hemaglobal*.

[The information on this sheet is produced for hypothetical discussion purposes only; and is not meant to bear any relationship to actual events.]